

Annex S1. Google Street View validation protocol

Overview

Coders review each sampled location in Google Street View (GSV) to determine whether cycling infrastructure is present around the follow-up date (approximately 1 January 2024) and the baseline date (approximately 1 January 2016). Only the cells in coloured columns are editable; all other cells contain formulas or identifiers.

Cycling-infrastructure definition: Cycling infrastructure is defined as a physically visible, dedicated facility for cycling. This includes (i) a dedicated cycleway alignment (corresponding to OSM highway=cycleway), or (ii) a cycle lane or track, whether on the carriageway or on the pavement (painted or kerb-separated), including contraflow lanes, corresponding to OSM cycleway* values in {lane, track, opposite_lane, opposite_track}.

Cycling is not coded as infrastructure when it is merely permitted or prioritised without a dedicated facility, for example bicycle streets, shared streets, pedestrian streets with bicycle access, traffic calming measures, or signage alone.

Data entry structure

Blue columns → main years (presence: 1/0/blank; month: 1-12)

Follow-up

- **2024** (pres24, mon24): fill if interpretable. If the view is unclear, or if the requested year is not available at that viewpoint, try moving the peg slightly along the same street to reveal 2024 imagery, but only to a point where the original location remains visible. If the peg cannot be moved and 2024 imagery cannot be interpreted, leave these cells blank.
- **2023** (pres23, mon23): same rule.

If both years remain uninterpretable (including cases where the peg cannot be moved or year selection is unavailable) → leave them blank and use the fallback year.

Baseline

- **2016** (pres16, mon16): fill if interpretable. If the view is unclear, or if the requested year is not available at that viewpoint, try moving the peg slightly along the same street to reveal 2016 imagery, but only to a point where the original location remains visible. If the peg cannot be moved or 2016 imagery cannot be interpreted, leave these cells blank.
- **2015** (pres15, mon15): same rule.

If both years remain uninterpretable (including cases where the peg cannot be moved or year selection is unavailable) → leave them blank and use the fallback year.

Yellow columns → fallback years

Used only when both main years for that period cannot be interpreted, even after a small peg adjustment (or when no adjustment is possible).

- **Follow-up fallback (2022):** pres22, mon22
- **Baseline fallback (2014):** pres14, mon14

If at least one main year is usable, leave fallback cells empty. If the fallback year is also not interpretable, leave fallback cells blank.

Green column → notes

Optional brief notes when something affects interpretation (peg movement, inability to move the peg, missing year options, blurred imagery, temporary works, ambiguous cases).